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## ABSTRACT

A SOT-MRAM layout, including a read active area and a write active area, a first word line and a second word line dividing the read active area into a first drain and a second drain at two sides and a first common source in the middle and dividing the write active area into a third drain and a fourth drain at two sides and a second common source in the middle, a read bit line connected with the first drain and second drain, a write bit line connected with the third drain and fourth drain, a MTJ connected on the first common source, and a SOT layer connected on the MTJ, with two ends connected respectively with the second common source and a source line.

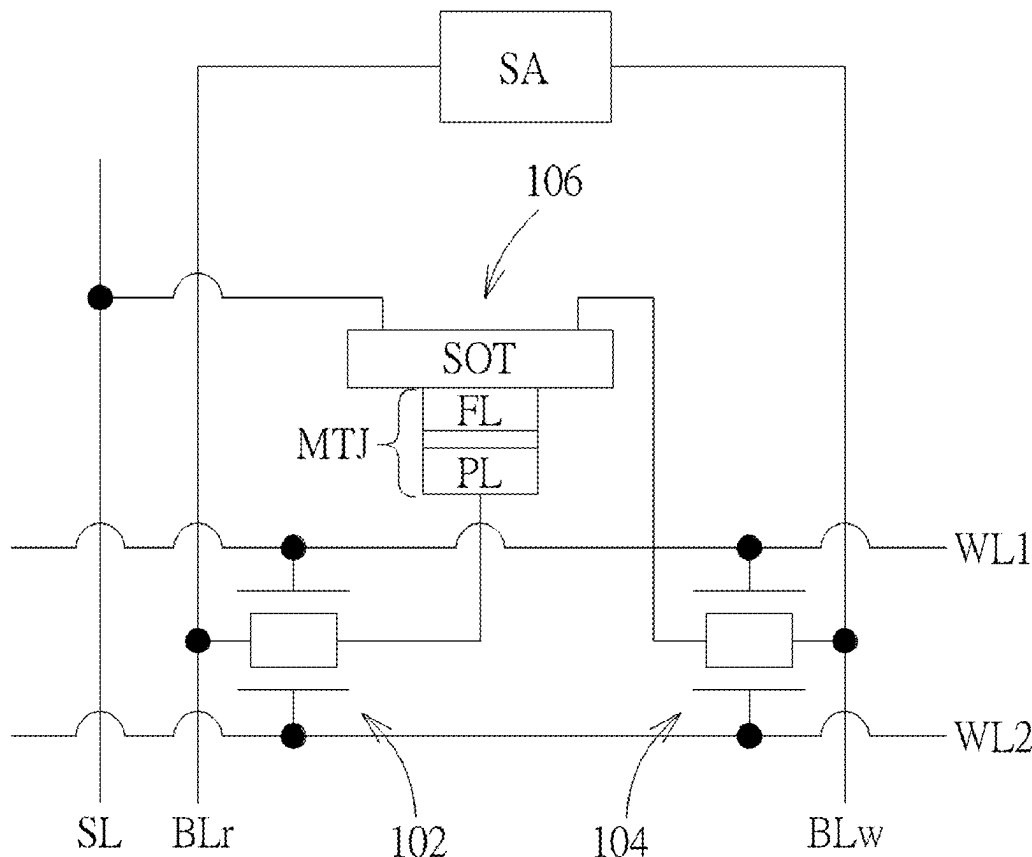
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Mar. 4, 2022 (CA) ..... 202210207577.4





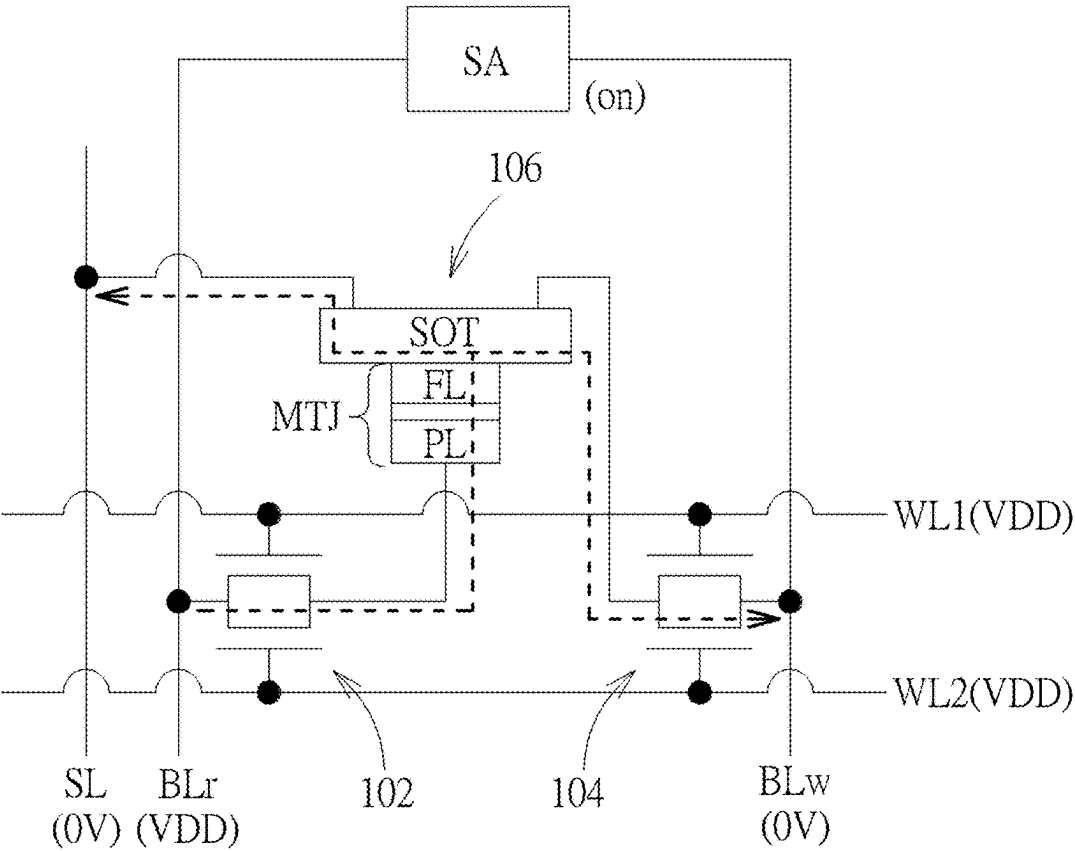


FIG. 3

|         | write 0 | write 1 | read |
|---------|---------|---------|------|
| WL1/WL2 | VDD     | VDD     | VDD  |
| BLw     | VDD     | 0       | 0    |
| BLr     | Hiz     | Hiz     | VDD  |
| SL      | 0       | VDD     | 0    |
| SA      | off     | off     | on   |

FIG. 4

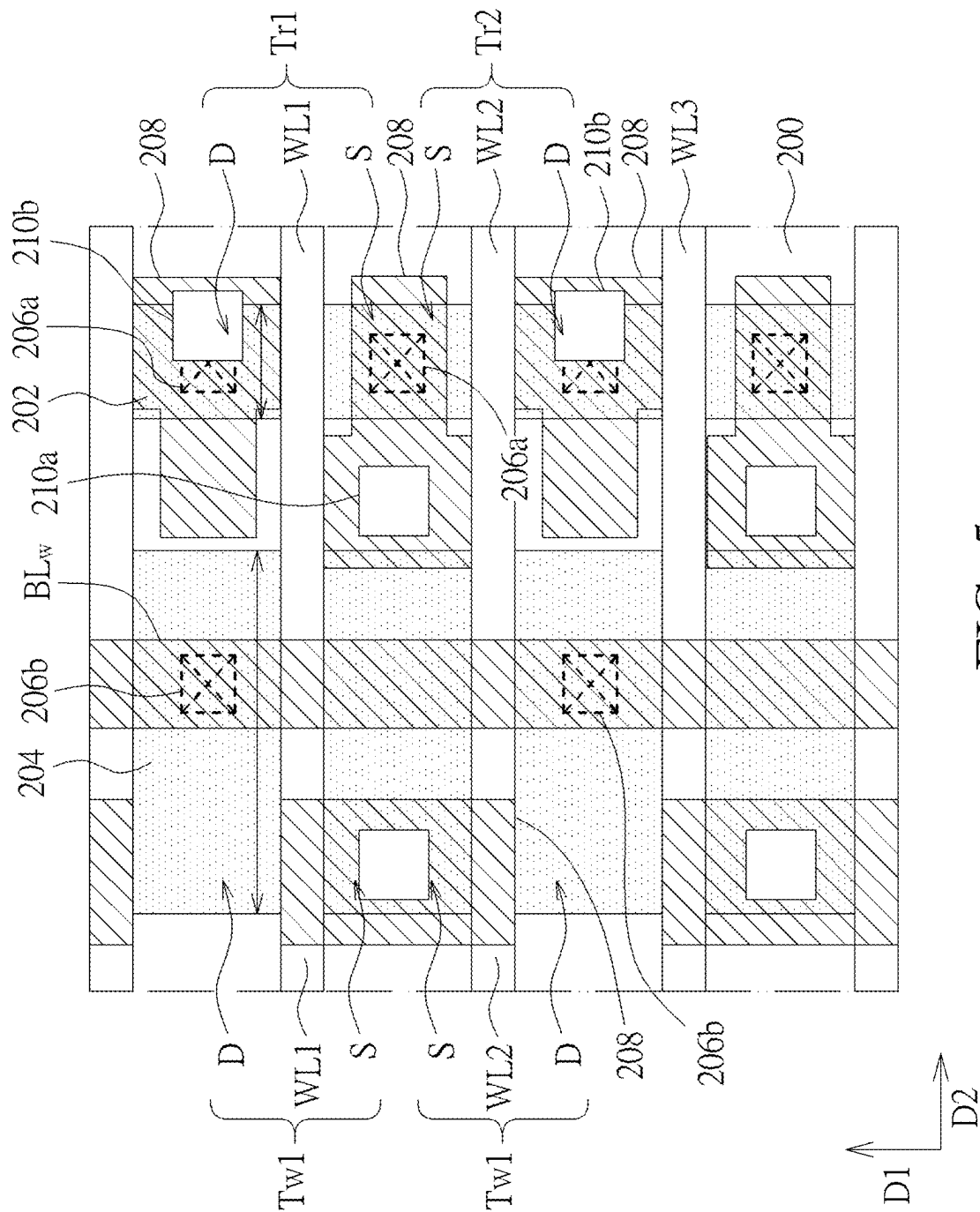


FIG. 5

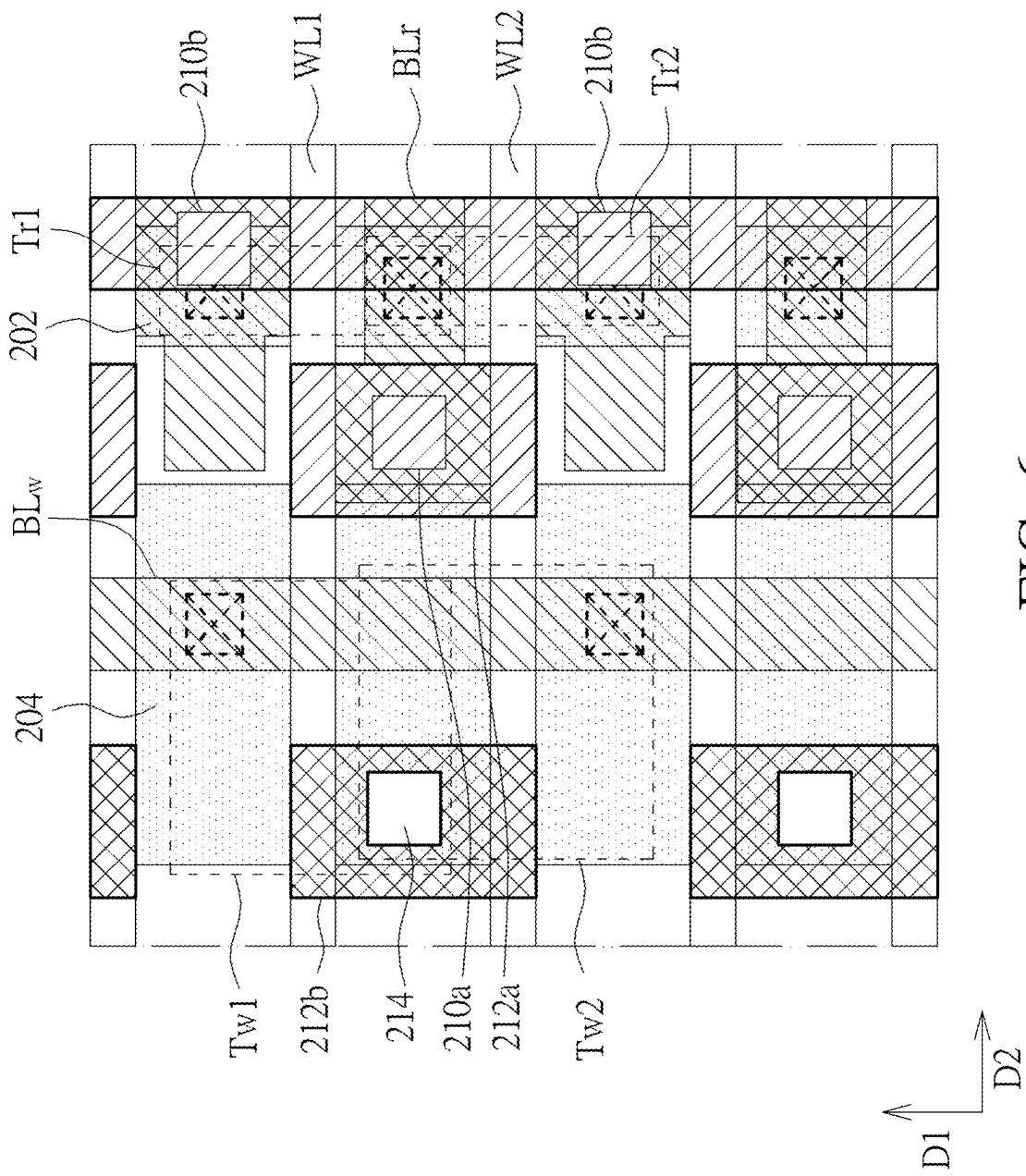


FIG. 6

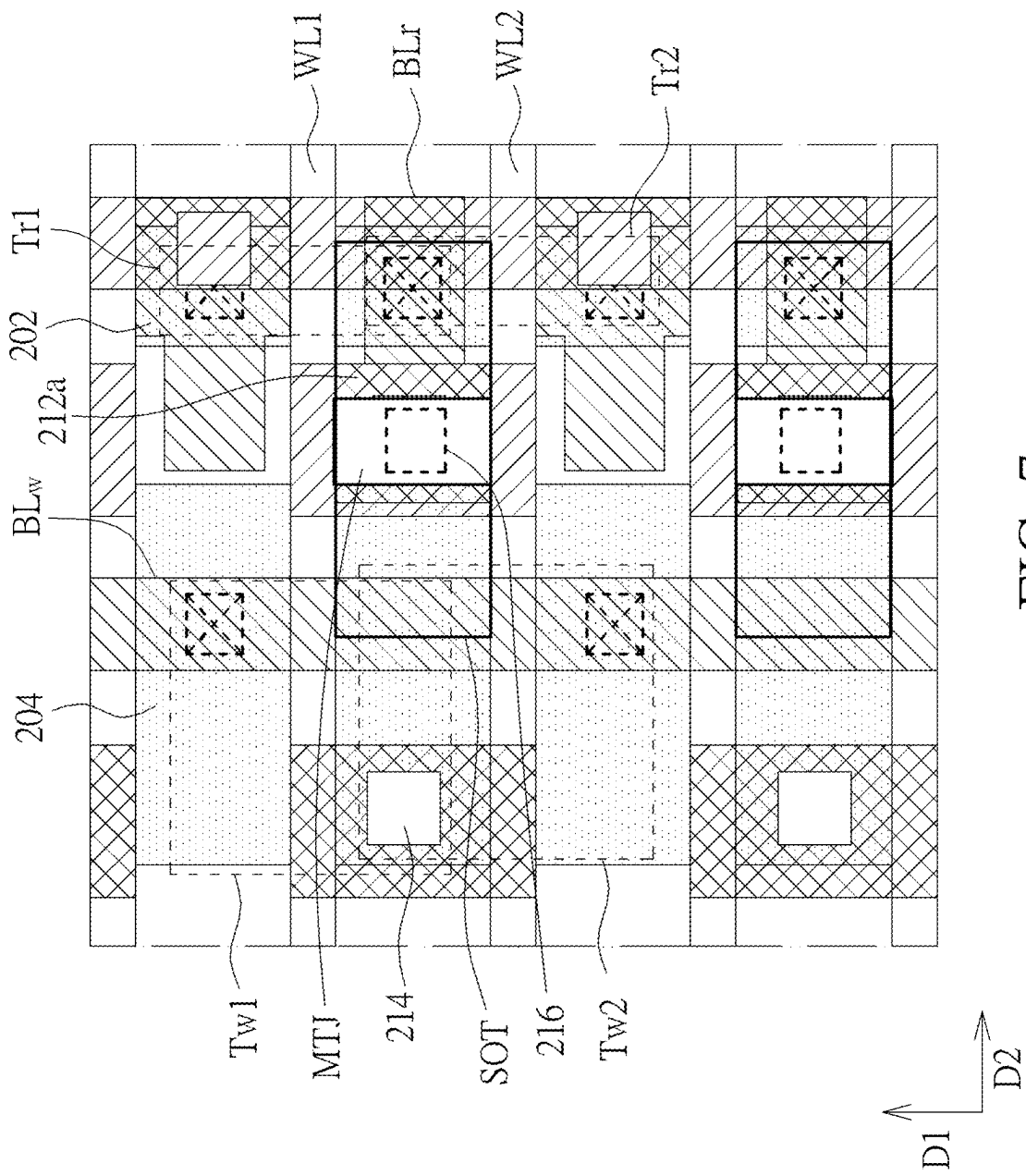


FIG. 7

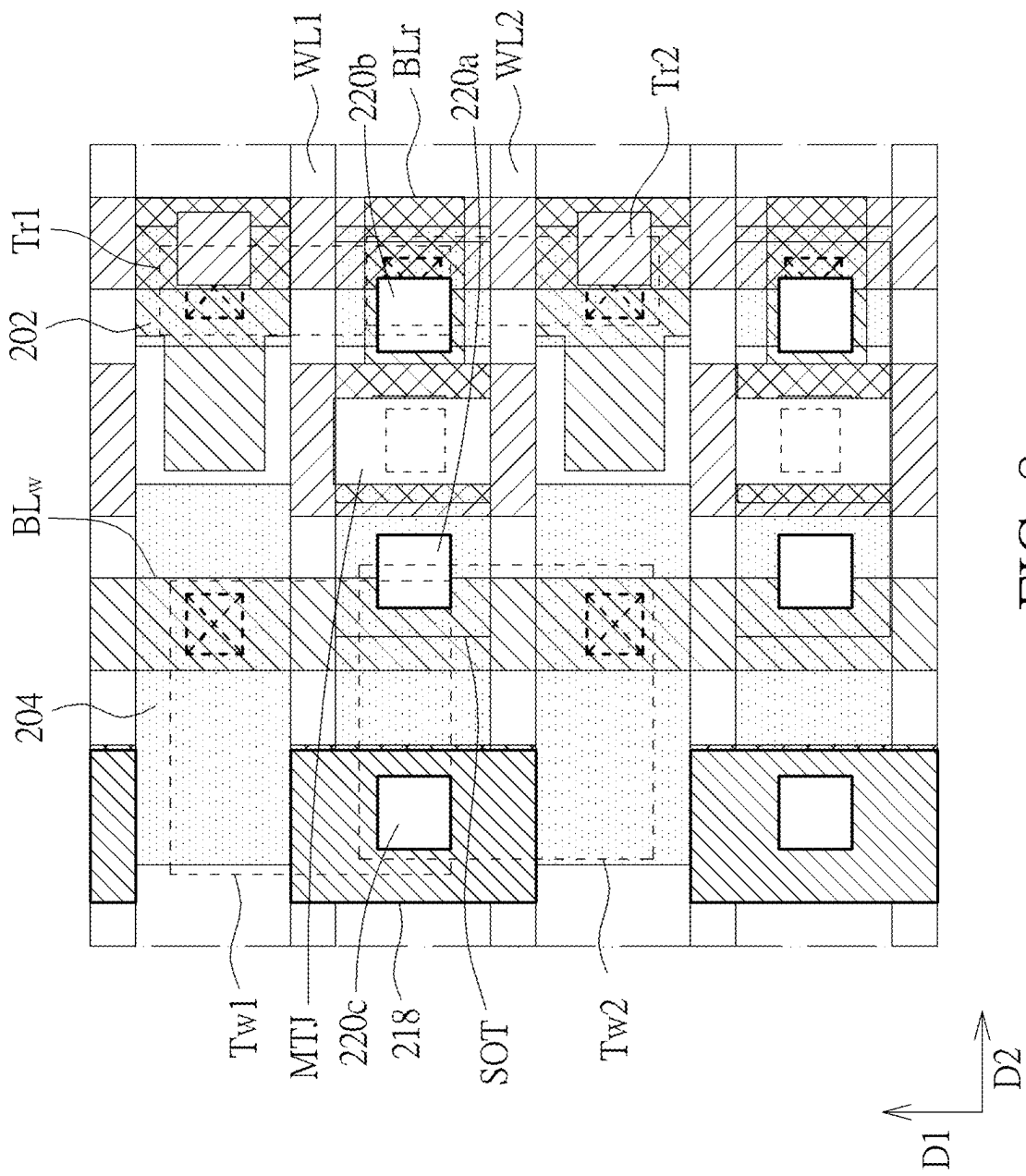


FIG. 8

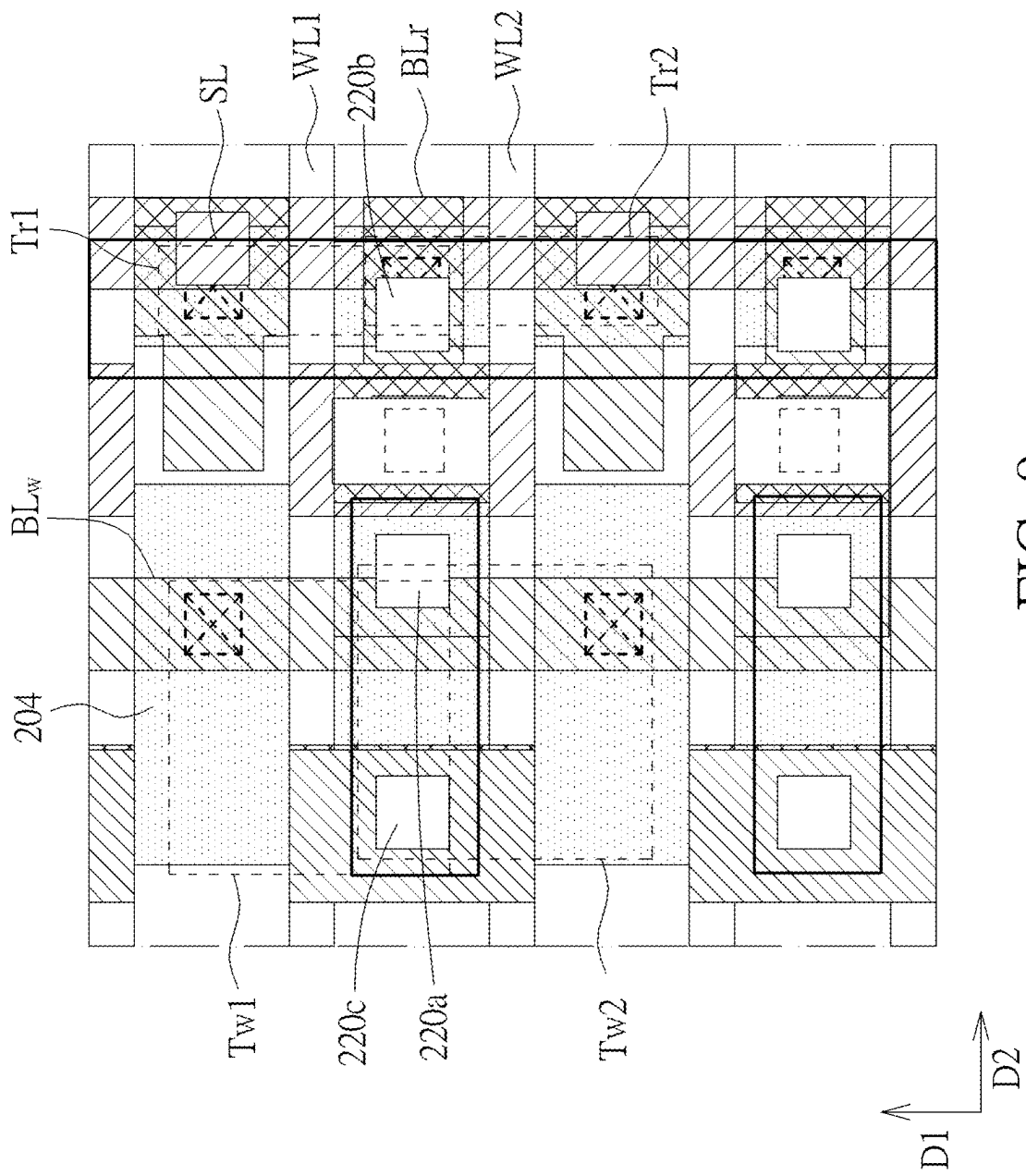


FIG. 9



## SPIN-ORBIT TORQUE MAGNETIC RANDOM ACCESS MEMORY CIRCUIT AND LAYOUT THEREOF

### CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a division of U.S. application Ser. No. 17/707,934, filed on Mar. 29, 2022. The content of the application is incorporated herein by reference.

### BACKGROUND OF THE INVENTION

#### 1. Field of the Invention

[0002] The present invention relates generally to a spin-orbit torque magnetic random access memory (SOT-MRAM) circuit and layout thereof, and more specifically, to a 4T1M (four transistors and one memory unit) SOT-MRAM circuit and layout thereof.

#### 2. Description of the Prior Art

[0003] Magnetic random access memory (MRAM) is considered to be a promising and emerging memory in the future of memory industry due to its fast read/write speeds, non-volatility and integratability with semiconductor process. Start from early magnetic field operation to recent commercially produced spin-transfer torque (STT) type, and further to current spin-transfer orbit (SOT) type researched and developed actively in the industry, the operation model of MRAM evolves correspondingly from indirectly application of magnetic field through current to directly granting temporal pulses for fast read/write operation.

[0004] As far as spin-orbit torque magnetic random access memory (SOT-MRAM) is concerned, since its architecture of read/write splitting three-terminals design, the current applied in write operation doesn't need to pass through magnetic tunnel junction, so that the tolerance and reliability of memory device may be improved. However, the adopting of three-terminal architecture results in more components required to be set on the layout area of the SOT-MRAM. Therefore, it is essential to improve the layout design in the development of SOT-MRAM technology in order to reduce required layout area. Regarding this, the approach of common word line or common bit line (read/write circuit lines shared by transistors) is adopted in conventional skill. Nevertheless, these approaches have existing disadvantages, for example, requiring additional isolating word lines or the gate widths of read/write active areas have to be the same. Besides, current SOT-MRAM design has quite low current efficiency. It may need to apply higher operating current in comparison to ordinary memory, which is also disadvantageous to the miniaturization of the device in the future. Therefore, those of skilled in the art still need to improve the layout structure of current SOT-MRAM in order to solve the issue of inefficient current usage due to the aforementioned structural design.

### SUMMARY OF THE INVENTION

[0005] In the light of the aforementioned disadvantages in conventional skills, the present invention hereby provides a novel spin-orbit torque magnetic random access memory (SOT-MRAM) circuit and layout thereof, with features of adopting 4T1M (four transistors and one memory unit) architecture having two read transistors and two write tran-

sistor connected in pair and in parallel, so that the memory unit may receive double operating current. In addition, the read/write active areas may have different widths and need not to form additional isolating word lines.

[0006] One aspect of the present invention is to provide a spin-orbit torque magnetic random access memory circuit, with structures including a read transistor pair having two read transistors in parallel, a write transistor pair having two write transistors in parallel, a spin-orbit torque magnetic random access memory unit with a magnetic tunnel junction and a spin-orbit torque layer, wherein one end of the magnetic tunnel junction is connected to a source of the read transistor pair and the other end of the magnetic tunnel junction is connected to the spin-orbit torque layer, and one end of the spin-orbit torque layer is connected to a source line and the other end of the spin-orbit torque layer is connected to a source of the write transistor pair, a read bit line connected to a drain of the read transistor pair, a write bit line connected to a drain of the write transistor pair, a first word line connected to a gate of one of the read transistor and a gate of one of the read transistor, and a second word line connected to a gate of another the read transistor and a gate of another the read transistor.

[0007] Another aspect of the present invention is to provide a spin-orbit torque magnetic random access memory layout, including a substrate with a read active area and a write active area formed thereon, a first word line and a second word line both extending through the read active area and the write active area and dividing the read active area into a first drain and a second drain at two sides and a first common source in the middle and dividing the write active area into a third drain and a fourth drain at two sides and a second common source in the middle, a read bit line on the read active area and connected with the first drain and the second drain, so that the first drain, the second drain, the first common source, the first word line and the second word line collectively constitute a read transistor pair having two read transistors in parallel, a write bit line on the write active area and connected with the third drain and the fourth drain, so that the third drain, the fourth drain, the second common source, the first word line and the second word line collectively constitute a write transistor pair having two write transistors in parallel, a magnetic tunnel junction on the first common source and connected with the first common source, and a spin-orbit torque layer on the magnetic tunnel junction and connected with the magnetic tunnel junction, and two ends of the spin-orbit torque layer are connected respectively with the second common source and a source line.

[0008] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

### BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings are included to provide a further understanding of the embodiments, and are incorporated in and constitute a part of this specification. The drawings illustrate some of the embodiments and, together with the description, serve to explain their principles. In the drawings:

**[0010]** FIG. 1 is a schematic circuit diagram of a spin-orbit torque magnetic random access memory (SOT-MRAM) in accordance with one preferred embodiment of the present invention;

**[0011]** FIG. 2 is a schematic circuit diagram of the SOT-MRAM in write operation in accordance with one preferred embodiment of the present invention;

**[0012]** FIG. 3 is a schematic circuit diagram of the SOT-MRAM in read operation in accordance with one preferred embodiment of the present invention;

**[0013]** FIG. 4 is a table list of applied voltages in different circuit lines in read operation and write operation of the SOT-MRAM in accordance with one preferred embodiment of the present invention; and

**[0014]** FIGS. 5-9 are layouts of the SOT-MRAM in different levels in accordance with one preferred embodiment of the present invention.

**[0015]** Relative dimensions and proportions of parts of the drawings have been shown exaggerated or reduced in size, for the sake of clarity and convenience in the drawings. The same reference signs are generally used to refer to corresponding or similar features in modified and different embodiments.

#### DETAILED DESCRIPTION

**[0016]** Reference will now be made in detail to exemplary embodiments of the invention, which are illustrated in the accompanying drawings in order to understand and implement the present disclosure and to realize the technical effect. It can be understood that the following description has been made only by way of example, but not to limit the present disclosure. Various embodiments of the present disclosure and various features in the embodiments that are not conflicted with each other can be combined and rearranged in various ways. Without departing from the spirit and scope of the present disclosure, modifications, equivalents, or improvements to the present disclosure are understandable to those skilled in the art and are intended to be encompassed within the scope of the present disclosure.

**[0017]** It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something). Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature relationship to another element(s) or feature(s) as illustrated in the figures.

**[0018]** As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon (Si), germanium (Ge), gallium arsenide (GaAs), indium phosphide (InP), etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

**[0019]** As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure, or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductor and contact layers (in which contacts, interconnect lines, and/or through holes are formed) and one or more dielectric layers.

**[0020]** In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. Additionally, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors, but may allow for the presence of other factors not necessarily expressly described, again depending at least in part on the context.

**[0021]** It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

**[0022]** First, please refer to FIG. 1, which is a schematic circuit diagram of a spin-orbit torque magnetic random access memory (SOT-MRAM) in accordance with one preferred embodiment of the present invention. The SOT-MRAM circuit of the present invention is composed of three main components, i.e. a read transistor pair **102**, a write transistor pair **104** and a SOT-MRAM unit **106**, which constitute the 4T1M (four transistors and one memory unit) MRAM architecture of the present invention, wherein the read transistor pair **102** consists of two read transistors connected in parallel and the write transistor pair **104** consists of two write transistors connected in parallel, and the SOT-MRAM unit **106** consists of a magnetic tunnel junction MTJ and a spin-orbit torque layer SOT. In the aspect of connection, one end of the magnetic tunnel junction MTJ of the SOT-MRAM unit **106** is connected to source of the read transistor pair **102** and the other end is contacted directly with the spin-orbit torque layer SOT. One end of the spin-orbit torque layer SOT is connected to a source line SL and the other end is connected to source of the write transistor pair **104**. The drain of read transistor pair **102** is connected to a read bit line BLr, and the drain of write transistor pair **104** is connected to a write bit line BLw. One of the gates of the read transistor pair **102** and one of the gates of the write transistor pair **104** are connected to a

common first word line WL1, and the other gate of the read transistor pair **102** and the other gate of the write transistor pair **104** are connected to a common second word line WL2. In addition, the read bit line BLr and the write bit line BLw are both connected to a sense amplifier SA.

**[0023]** In the aspect of structure, the magnetic tunnel junction MTJ may consist of one pinned layer PL and one free layer FL. The material of pinned layer PL may be ferromagnetic material, which includes but not limited to iron (Fe), cobalt (Co), nickel (Ni) or the alloy thereof like CoFeB or CoFe. Alternatively, the pinned layer PL may be made of antiferromagnetic (AFM) material, which includes but not limited to FeMn, PtMn, IrMn, NiO or the combination thereof. The function of pinned layer PL is to fix or limit the directions of magnetic moment of adjacent layer structures, so that they will not be flipped by external magnetic field. The material of free layer FL may be ferromagnetic material, which includes but not limited to Fe, Co, Ni or the alloy thereof like CoFeB or CoFe. The magnetic moment of the free layer FL may be freely flipped and altered through external electric field. When the magnetization directions of the pinned layer PL and the free layer **114** are the same, the spin directions of a majority of electrons in the two ferromagnetic materials are the same, so that they will have higher tunneling probability, thereby having larger tunneling current and the magnetic tunnel junction MTJ will be rendered in low-resistance state (“0” bit). Conversely, the magnetic tunnel junction MTJ will be rendered in high-resistance state (“1” bit). Furthermore, the material of spin-orbit torque layer SOT may include tungsten (W). In the present invention, spin moment may be generated by spin Hall effect (SHE) when current flows through the spin-orbit torque layer SOT, so as to alter the magnetic moment of magnetic materials in the magnetic tunnel junction MTJ. This is principle how to achieve data storage by MRAM through the approach of altering the spin direction of electrons to change the electrical resistance.

**[0024]** Please refer now to FIG. 2, which is a schematic circuit diagram of the SOT-MRAM in the write operation in accordance with one preferred embodiment of the present invention, and may also refer collectively to the table list of applied voltages in different circuit lines in the write operation of the SOT-MRAM in FIG. 4. In the operation of writing low logic state “0” data in the SOT-MRAM of the present invention, an operating voltage VDD is applied from the first word line WL1 and the second word line WL2 to open the transistors in the read transistor pair **102** and the write transistor pair **104**, and an operating voltage VDD is applied from the write bit line BLw to the grounded source line SL (OV) sequentially through the opened write transistor pair **104** and the spin-orbit torque layer SOT, while the read bit line BLr is in high-resistance state (Hiz). At this time, no current will flow through the magnetic tunnel junction MTJ and the read bit line BLr. The applied current will alter the magnetic moment of magnetic materials in the magnetic tunnel junction MTJ in the aforementioned process of flowing through the spin-orbit torque layer SOT and render it in low-resistance state (“0” bit). The sense amplifier SA in this write process is in OFF state. This is the operation of writing low logic state “0” data in the present invention.

**[0025]** On the other hand, refer still to FIG. 2 and FIG. 4. In the operation of writing high logic state “1” data in the SOT-MRAM of the present invention, similarly, an operating voltage VDD is applied from the first word line WL1 and

the second word line WL2 to open the transistors in the read transistor pair **102** and the write transistor pair **104**, but an operating voltage VDD is applied instead from the source line SL to the grounded write bit line BLw (OV) sequentially through the spin-orbit torque layer SOT and the opened write transistor pair **104**, while the read bit line BLr is in high-resistance state (Hiz). At this time, no current will flow through the magnetic tunnel junction MTJ and the read bit line BLr. The applied current will alter the magnetic moment of magnetic materials in the magnetic tunnel junction MTJ in the aforementioned process of flowing through the spin-orbit torque layer SOT and render it in high-resistance state (“1” bit). The sense amplifier SA in this write process is in OFF state. This is the operation of writing high logic state “1” data in the present invention.

**[0026]** Please refer now to FIG. 3, which is a schematic circuit diagram of the SOT-MRAM in the read operation in accordance with one preferred embodiment of the present invention, and may also refer collectively to the table list of applied voltages in different circuit lines in the read operation of the SOT-MRAM in FIG. 4. In the read operation of the SOT-MRAM of the present invention, an operating voltage VDD is applied from the first word line WL1 and the second word line WL2 to open the transistors in the read transistor pair **102** and the write transistor pair **104**, and an operating voltage VDD is applied from the read bit line BLr to the grounded source line SL and write bit line BLw (OV) sequentially through the opened read transistor pair **102**, the magnetic tunnel junction MTJ and the spin-orbit torque layer SOT. In this process, the sense amplifier SA will be turn ON so that it may receive currents from the read bit line BLr and the write bit line BLw. The resistance of the magnetic tunnel junction MTJ may be obtained through the measured current value, thereby further obtaining its logic state.

**[0027]** After explaining the circuit structure of the present invention, the following embodiments will explain layouts of the SOT-MRAM in different levels in accordance with one preferred embodiment of the present invention, which may help readers further understand the relative positions and connections of the components in the SOT-MRAM structure of present invention.

**[0028]** Please refer first to FIG. 5. The SOT-MRAM of present invention is manufactured on a substrate **200**. The substrate **200** may include a wide array of semiconductor materials, such as silicon (Si), germanium (Ge), gallium arsenide (GaAs), indium phosphide (InP), but not limited thereto. A read active area **202** and a write active area **204** are formed on the substrate **200** through doping dopants into the substrate **200** by ion implantation processes. As shown in FIG. 5, in the preferred embodiment of the present invention, the read active area **202** and the write active area **204** extend in a first direction D1 and space apart from each other at a distance, and since the write operation of the MRAM requires larger current, the width w1 of the write active area **204** in a second direction D2 perpendicular to the first direction D1 is designedly larger than the width w2 of the read active area **202** in the second direction D2. Multiple word lines WL1~WL3 are set on the substrate **200**, extend in the second direction D2 and space apart from each other. The word lines WL1~WL3 extend across the read active area **202** and the write active area **204**, wherein each word line WL1~WL3 and the passed read active areas **202** at two sides of it constitute a read transistor, and each word line

WL1~WL3 and the passed write active areas **204** at two sides of it constitute a write transistor.

[0029] For example, the word line WL1 and the source S and drain D at two sides of the read active area **202** constitute a read transistor Tr1, the word line WL2 and the source S and drain D at two sides of the read active area **202** constitute a read transistor Tr2, the word line WL1 and the source S and drain D at two sides of the write active area **204** constitute a write transistor Tw1, and the word line WL2 and the source S and drain D at two sides of the write active area **204** constitute a write transistor Tw2. The word lines WL1, WL2 function as gates of the aforementioned transistors. It can be seen in the figure that the read transistors Tr1, Tr2 share a source S, and their drains are also connected to a common read bit line above (will be described and shown in figures later) through interconnect structures like contacts and metal interconnection layers. Therefore, in the embodiment of present invention, the read transistors Tr1, Tr2 are connected in parallel into a read transistor pair (i.e. **102** in FIG. 1). Similarly, the write transistors Tw1, Tw2 share a source S, and their drains are also connected to a common write bit line BLw above through contacts **206b**. Therefore, in the embodiment of present invention, the write transistors Tw1, Tw2 are connected in parallel into a write transistor pair (i.e. **104** in FIG. 1).

[0030] Refer still to FIG. 5. What above the word lines WL1~WL3 are the metal interconnect formed structures in CMOS BEOL (back-end-of-line) process, such as contacts **206a**, **206b**, first metal layer (M1) **208** and vias **210a**~**210c**. In the embodiment, the sources S and drains D of the read transistors Tr1, Tr2 are all connected to a first metal layer (metal island) **208** above through contacts **206a**. The sources S of write transistors Tw1, Tw2 are also connected to the first metal layers **208** above through contacts (which overlap the patterns of vias **210c** formed thereon), however, their drains are connected to a common write bit line BLw above through contacts **206b**. Vias are set on the aforementioned first metal layer **208** to further connect the structures in upper levels, such as second metal layer (M2). It can be seen in the figure that vias **210a**, **210b** on the read transistors Tr1, Tr2 are in staggered arrangement in the first direction D1, wherein the vias **210b** for connecting the drain D overlap the underlying read active area **202**, while the vias **210a** for connecting the source S are set at positions between the read active area **202** and write active area **204**. In this embodiment, the write bit line BLw are also in the level of first metal layer **208**, and extends in the first direction D1 and overlap the underlying write active area **204**.

[0031] Please refer now to FIG. 6. The aforementioned vias (V1) **210a**~**210c** will be connected to overlying second metal layer (M2, metal islands) or the read bit line. As shown in FIG. 6, the via **210a** on the common source of the read transistors Tr1, Tr2 connects to a second metal layer **212a** above, while the via **210b** on the drain connects to a common read bit line BLr above, so as to achieve the effect of connecting the read transistors Tr1, Tr2 in parallel. The via (which overlaps the pattern of via **214** thereon) on the common source of the write transistors Tw1, Tw2 connects to a second metal layer **212b** above. In the embodiment of present invention, the read bit line BLr extends in the first direction and partially overlaps the read active area **202** below, and it is in the same level as the second metal layers **212a**, **212b**. The second metal layer **212a** is roughly between the read active area **202** and the write active area **204**, the

second metal layer **212b** roughly overlaps the write active area **204**, and the second metal layer **212a** and the second metal layer **212b** are symmetric in the second direction D2 with respect to the write bit line BLw below, wherein via **214** is set on the second metal layer **212b** to further connect to structures in upper layers, such as the third metal layer (M3).

[0032] Please refer now to FIG. 7. With regard to the read transistors Tr1, Tr2, the common source of read transistors Tr1, Tr2 is connected to the second metal layer **212a** (FIG. 6) sequentially through the contact **206a**, the first metal layer **208** and the via **210a** (FIG. 5). Structures like tungsten via **216**, magnetic tunnel junction MTJ and spin-orbit torque layer SOT are set sequentially on the second metal layer **212a**. The magnetic tunnel junction MTJ is connected with the second metal layer **212a** below through the tungsten via **216** and directly contacts the overlying spin-orbit torque layer SOT, preferably the middle section of the spin-orbit torque layer SOT, so as to constitute the SOT-MRAM unit (i.e. **106** in FIG. 1) of the present invention. Furthermore, the magnetic tunnel junction MTJ is preferably set at a position between the read active area **202** and the write active area **204**, while the spin-orbit torque layer SOT is set at a position between the word lines WL1, WL2 and extends in the second direction D2 to partially overlap the parts like read active area **202**, write active area **204**, read bit line BLr and write bit line BLw. In general, the aforementioned tungsten via **216**, magnetic tunnel junction MTJ and spin-orbit torque layer SOT may be in the same level as the one of via **214**.

[0033] Please refer now to FIG. 8. Following the spin-orbit torque layer SOT in previous figure, vias **220a**, **220b** are set respectively on two ends of the spin-orbit torque layer SOT and are roughly symmetric in the second direction D2 with respect to the magnetic tunnel junction MTJ below, wherein the via **220a** overlaps the write bit line BLw and the write active area **204** below, while the via **220b** overlaps the read bit line BLr and the read active area **202** below. With regard to the write transistors Tw1, Tw2, the via **214** (FIG. 7) is connected to a third metal layer (M3) **218** above, wherein its plan pattern may completely overlap the underlying second metal layer **212b** (FIG. 7). Another via **220c** is further set on the third metal layer **218**, wherein its plan pattern may completely overlap the via **214** (FIG. 7) and the via **210c** (FIG. 5) below. The aforementioned vias **220a**, **220b**, **220c** are all preferably set at positions between the word lines WL1, WL2. It can be seen in the figure that the read bit line BLr and the write bit line BLw are preferably symmetric in the second direction D2 with respect to the magnetic tunnel junction MTJ.

[0034] Lastly, please refer to FIG. 9. Following the aforementioned vias **220a**, **220b**, **220c**, the via **220b** on one end of the spin-orbit torque layer SOT connects with a source line SL above. In the embodiment of present invention, the source line SL extends in the first direction D1 and overlaps the read bit line BLr and the read active area **202** (FIG. 5) below. The via **220a** on another end of the spin-orbit torque layer SOT connects with a bridge part **222** in an upper layer and is further connected to the via **220c** of the write active area **204** through the bridge part **222**. In the embodiment of present invention, the bridge part **222** is set at a position between the word lines WL1, WL2 and extends in the second direction D2, and it also roughly overlaps the write active area **204** and partially overlaps the spin-orbit torque layer SOT below.

[0035] Refer still to FIG. 9. It can be clearly understood through the aforementioned layout patterns that, in the embodiment of present invention, one end of the spin-orbit torque layer SOT of the SOT-MRAM is connected to the source line SL to form one terminal of the memory device, the other end of the spin-orbit torque layer SOT is connected to the write transistor pair and the write bit line BLW connected therewith to form another terminal of the memory device. Furthermore, the SOT-MRAM unit is connected downwardly to the read transistor pair and the read bit BLr line connected therewith to form the third terminal of the memory device.

[0036] In summary to the aforementioned embodiments, the SOT-MRAM circuit and layout thereof provided in the present invention features the 4T1M (four transistors and one memory unit) architecture, including a read transistor pair with two read transistors connected in parallel and a write transistor pair with two write transistors connected in parallel, so that required layout area may be reduced in the manner of sharing common drains. Furthermore, the read transistor pair and the write transistor pair are both provided with two drains connected to a common bit line for providing operating current, so that the memory unit may receive double operating current in comparison to the one in conventional skill, thereby solving the issue of inefficient current usage of the MRAM. In addition, the read active area and the write active area may have different width to provide design flexibility for layout and need not to form additional isolating word lines for isolating memory units.

[0037] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. A spin-orbit torque magnetic random access memory layout, comprising:

- a substrate with a read active area and a write active area formed thereon;
- a first word line and a second word line, both extending through said read active area and said write active area and dividing said read active area into a first drain and a second drain at two sides and a first common source in the middle and dividing said write active area into a third drain and a fourth drain at two sides and a second common source in the middle;
- a read bit line on said read active area and connected with said first drain and said second drain, so that said first drain, said second drain, said first common source, said first word line and said second word line collectively constitute a read transistor pair comprising two read transistors in parallel;
- a write bit line on said write active area and connected with said third drain and said fourth drain, so that said third drain, said fourth drain, said second common source, said first word line and said second word line collectively constitute a write transistor pair comprising two write transistors in parallel;
- a magnetic tunnel junction on said first common source and connected with said first common source; and
- a spin-orbit torque layer on said magnetic tunnel junction and connected with said magnetic tunnel junction, and

two ends of said spin-orbit torque layer are connected respectively with said second common source and a source line.

2. The spin-orbit torque magnetic random access memory layout of claim 1, wherein said read active area, said write active area, said read bit line, said write bit line and said source line extend in a first direction, and said first word line and said second word line extend in a second direction perpendicular to said first direction.

3. The spin-orbit torque magnetic random access memory layout of claim 1, wherein said write bit line is in a first metal layer of a metal interconnection structure and completely overlaps underlying said write active area, and said write bit line is connected with said third drain and said fourth drain through contacts.

4. The spin-orbit torque magnetic random access memory layout of claim 1, wherein said read bit line is in a second metal layer of a metal interconnection structure and partially overlaps underlying said read active area, and said read bit line is connected with said third drain and said fourth drain through contacts, a first metal layer and first vias.

5. The spin-orbit torque magnetic random access memory layout of claim 1, wherein said magnetic tunnel junction is in a level between a second metal layer and a third metal layer of a metal interconnection structure and is at a position between said read active area and said write active area, and said magnetic tunnel junction is connected with said first common source through said second metal layer, a first via, a first metal layer and a contact.

6. The spin-orbit torque magnetic random access memory layout of claim 1, wherein said spin-orbit torque layer is in a level between a second metal layer and a third metal layer of a metal interconnection structure and a middle part of said spin-orbit torque layer is connected with said magnetic tunnel junction below.

7. The spin-orbit torque magnetic random access memory layout of claim 6, wherein said first word line and said second word line extend in a second direction, and said spin-orbit torque layer, said write bit line and said read bit line are symmetric in said second direction with respect to said magnetic tunnel junction, and one end of said spin-orbit torque layer partially overlap said read bit line and said first common source, and another end of said spin-orbit torque layer partially overlap said write bit line and said second common source.

8. The spin-orbit torque magnetic random access memory layout of claim 7, wherein said spin-orbit torque layer is between said first word line and said second word line.

9. The spin-orbit torque magnetic random access memory layout of claim 1, wherein said source line is in a fourth metal layer of a metal interconnection structure and partially overlap said read bit line, said read active area and said spin-orbit torque layer below, and said source line is connected with one end of said spin-orbit torque layer below through a third via.

10. The spin-orbit torque magnetic random access memory layout of claim 9, wherein another end of said spin-orbit torque layer is connected to a bridge part in said fourth metal layer above through another third via and further connected downwardly to said second common source through said bridge part.

11. The spin-orbit torque magnetic random access memory layout of claim 10, wherein said bridge part is

above said second common source between said first word line and said second word line and partially overlaps said spin-orbit torque layer.

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